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Case Study of Iraq's Longest Horizontal Well and Data-Driven Optimization of Drilling Parameters Using Multi-Objective Optimization

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Abstract

This paper presents a field case study of EBST-26-2H, the longest horizontal well drilled in Iraq to date, located in the East Baghdad South (EBS) oilfield. The well reached a total measured depth of 5,449.6 m, including a 2,826 m horizontal section in the Upper Cretaceous Tanuma carbonate reservoir. Drilled using a 6" bit and 3-1/2" drill pipe under extreme downhole conditions, the objective was to share drilling practices and engineering innovations that enabled successful execution of the extended horizontal section through full equipment utilization, precise well design, and innovative field operations—addressing key challenges related to friction, torque, hydraulics, and directional control while optimizing operational cost.

To tackle the challenges of ultra-long horizontal drilling with small-diameter tools, the following approaches were adopted: Friction and torque were mitigated through optimized BHA design and a high-lubricity, low-density water-based mud (WBM) system enhanced with both solid and liquid lubricants, lowering the lateral friction coefficient to ~0.25. Hydraulic pressure limitations were managed by upgrading surface equipment (high-pressure pumps and manifolds) and reducing pump pressure through optimized mud weight and controlled flow rate; effectiveness was validated via real-time monitoring and periodic short trips. Drillstring vibration and RSS malfunction were resolved by optimizing drilling parameters (e.g., increasing rotary speed to 110-120 RPM, moderating weight on bit); RSS issues were addressed by increasing pump rate to wash out debris and executing 400 m short trips, successfully restoring steering functionality.

The operation was completed 7 days ahead of schedule, validating the success of the proposed techniques: Friction reduction enabled effective weight transfer without string buckling or joint failure. Hydraulics were

managed within pressure limits, achieving sufficient hole cleaning with no significant pack-off or stuck-pipe events. Directional control was re-established post-RSS malfunction, keeping the well on planned trajectory without sidetracking. Completion and testing were successful, with open-hole completion and post-acidizing production exceeding 11,000 BOPD—establishing a benchmark for ERD wells in Iraq.

This paper provides the first documented case of drilling a record-length horizontal well in Iraq using standard 3-1/2" drill pipe and WBM, demonstrating a cost-effective approach to extended-reach drilling that fully utilizes existing equipment. It also introduces a data-driven optimization perspective using a multi-objective NSGA-II model to recommend future drilling parameters (weight on bit, rotary speed, flow rate) based on field data—bridging practical engineering with machine learning to enhance well performance. These results demonstrate that with rigorous engineering and operational adaptability, record-length horizontal wells can be drilled cost-effectively using conventional equipment.

Introduction

The East Baghdad South (EBS) oilfield is located in the eastern suburbs of Baghdad, Iraq. As the city expanded, drilling operations in this urban field faced increasingly complex surface constraints due to dense population, sensitive land ownership issues, historical heritage zones, and vital infrastructure. Such limitations make wellsite acquisition and construction extremely difficult. Urban oilfields in Iraq and the broader Middle East are encountering similar restrictions on surface access, which has accelerated the adoption of long horizontal wells and cluster drilling strategies to maximize subsurface development from limited sites [1]. In response to these challenges, horizontal drilling technology with extended reach laterals has become a focal point for field development.

Historically, horizontal wells in the EBS field have lateral lengths of about 1,200-2,000 m. The EBST-26-2H well was designed as a pilot to test the feasibility of drilling a much longer horizontal section under constrained surface conditions using only conventional equipment. This well pushed the limits of standard 3-1/2" drill pipe and a water-based mud (WBM) system to drill an extended-reach lateral, since higher-spec assets (e.g. 4" drill pipe or synthetic/oil-based mud) were unavailable due to logistics and environmental regulations [2]. The target reservoir, the Upper Cretaceous Tanuma formation, has low permeability but is a brittle, competent carbonate-characteristics which made it suitable for a long open-hole horizontal completion without requiring specialized mud for wellbore stability [3].

The well reached a measured depth (MD) of 5,449.6 m, including a 2,826 m horizontal section through the Tanuma reservoir. This set a new record for the longest horizontal section drilled in Iraq [4]. Until recently, most horizontal wells in Iraq were limited to less than 2,000 m laterals, and few attempts were made at extended-reach drilling due to lack of necessity and suitable infrastructure [5]. The success of EBST-26-2H under standard rig capabilities demonstrates that ultra-long horizontals are achievable in Iraq's fields, even under surface and equipment constraints.

This paper presents the engineering design, field execution, and operational adjustments that enabled the successful delivery of EBST-26-2H. Key drilling challenges - including high friction and torque, hydraulic limitations, severe downhole vibration leading to rotary steerable system (RSS) malfunctions, and other secondary issues - were encountered and mitigated through a combination of optimized bottom-hole assembly (BHA) design, mud system enhancements, controlled drilling practices, and on-the-fly contingency measures. In particular, multiple RSS tool failures were handled without significant non-productive time, and a robust contingency workflow was developed for future extended-reach cluster drilling. As a forward-looking extension, a data-driven post-drill analysis using a multi-objective genetic algorithm (NSGA-II) was implemented to optimize drilling parameters (weight on bit, rotary speed, flow rate) for maximizing penetration rate while suppressing stick-slip vibrations. By combining field results with machine learning and evolutionary optimization, the study provides insights to guide future wells in similar environments. The following sections detail the geological setting and well planning, drilling challenges

and solutions deployed, the RSS malfunction events and remedies, the contingency protocols established, and the multi-objective optimization methodology and results.

Field and Geological Overview

EBST-26-2H is a horizontal development well in the East Baghdad South field (Block S2), targeting the Upper Cretaceous Tanuma carbonate reservoir. It was drilled as an extended-reach offset to earlier horizontal wells (e.g., EBST-9-2H, EBST-9-3H) to maximize reservoir contact from a limited surface location. Each well in this program was planned to reach roughly 3.8 km MD with ~1.2 km of horizontal hole through the Tanuma pay zone. The Tanuma formation provides about 11 m of net oil pay (subzones T1-T4) in this area, and the horizontal section was steered within these upper Tanuma limestone layers to maximize exposure while avoiding shaley intervals. The 6" open-hole completion across the reservoir (i.e. no liner run in the horizontal) was chosen to minimize skin and facilitate production.

Geologically, the EBS Field is a complex, faulted anticline with multiple stacked reservoirs (Hartha, Sa'adi, Tanuma, Khasib, Zubair, etc.). The Tanuma formation is an Upper Cretaceous carbonate ~100-110 m thick in Block S2. Its upper portion consists of porous oil-bearing limestones (the target of the horizontals), while the lower Tanuma includes marls and shales. The overlying Shiranish formation (also Upper Cretaceous) is a marly limestone with interbedded shales, and above that lie Tertiary and younger formations such as the Lower Fars (Middle Miocene) - a 400+ m sequence of interbedded anhydrite and claystone - and other clastics. This stratigraphy presented several drilling hazards that were identified from nearby wells and incorporated into the planning of EBST-26-2H and its sister wells.

Offset Well Challenges: Previous wells in the area highlighted key issues related to the geology and were instructive for EBST-26-2H's design.

Lost Circulation: Severe mud losses were common in certain formations. For example, offset wells EBS-Z5 and EBS-H1-2-9H experienced partial to total losses in the Upper Cretaceous Nahr Umr sandstone and Jeribe limestone, and well EBS-Z12 lost ~800 m³ of mud in the Shuaiba formation. These thief zones (including fractured carbonates in formations like the Aaliji and Shiranish) readily absorbed drilling fluid. To mitigate this, the mud program for EBST-26-2H was proactive: the WBM was pre-treated with lost circulation material (LCM) prior to entering high-risk zones, and mud weight was kept at the lower end of the safe window to minimize overbalance. Equivalent circulating density (ECD) was closely managed by controlling mud rheology and circulation rates to avoid fracturing the formation. Contingency plans (which were later implemented) included spotting LCM pills and, if necessary, setting cement plugs to seal severe loss zones. Sufficient LCM and cement were kept on site, and the chosen reservoir drill-in fluid was formulated with bridging agents so it could tolerate minor losses without compromising wellbore stability.

Hydrogen Sulfide (H₂S): The EBS field is partially sour, with certain reservoirs containing high H₂S levels. Offsets drilling the underlying Khasib formation recorded extremely high H₂S in produced gas (6,000-10,000 ppm range). Although Tanuma itself was not expected to be very sour, strict H₂S safety measures were mandatory. Continuous gas monitoring was employed from the moment drilling proceeded below surface casing. All personnel were H₂S-trained and equipped with personal monitors and escape packs. The mud was conditioned to a high pH (~9.5+) to reduce H₂S solubility, and an H₂S scavenger was added to chemically neutralize any H₂S encountered. These precautions protected the crew and equipment (preventing sulfide stress cracking of tubulars). Fortunately, EBST-26-2H itself encountered only trace H₂S, but the preparedness was crucial for safe operations.

Wellbore Stability: Maintaining hole stability was a concern given the interbedded weak formations and pressure transitions. The Lower Fars (above Tanuma) consists of anhydrite-claystone layers prone to creep and swelling, often causing tight hole or stuck pipe if drilling pauses too long. The drilling plan called for frequent reaming while drilling Lower Fars and regular wiper trips to keep the hole in gauge. Static time in open hole was minimized (e.g. avoid staying stationary in one spot for more than a few minutes) to reduce

sticking risk. Deeper in the curve and lateral, the Tanuma and Shiranish shales (interbedded with carbonates) are water-sensitive and prone to sloughing when exposed to water-based mud. To improve stability, the mud weight was maintained sufficiently high (within safe limits) to support the formation, and mud filtrate loss was controlled to <3 mL (API) to minimize shale hydration. Inhibitive additives like KCl and asphaltic resins were used in the reservoir drill-in fluid to strengthen the wellbore and inhibit clay swelling. Operationally, high-viscosity sweeps and periodic short trips (every few hundred meters in the lateral) were scheduled to clear cuttings and prevent bed build-up. By actively managing mud chemistry and drilling practices, the team aimed to avoid stuck pipe or collapse incidents through these troublesome intervals.

In summary, the geological and offset well review shaped a drilling program that anticipated major challenges. Preventative measures for lost circulation, sour gas, and wellbore instability were integrated into the plan for EBST-26-2H. The well's slimhole design (6" hole with 3-1/2" drill pipe) further required careful trajectory control to limit doglegs (planned $<4.5^\circ/30$ m in the lateral) and ensure a smooth wellpath, reducing friction and enabling eventual completion operations (e.g. running coiled tubing for stimulation).

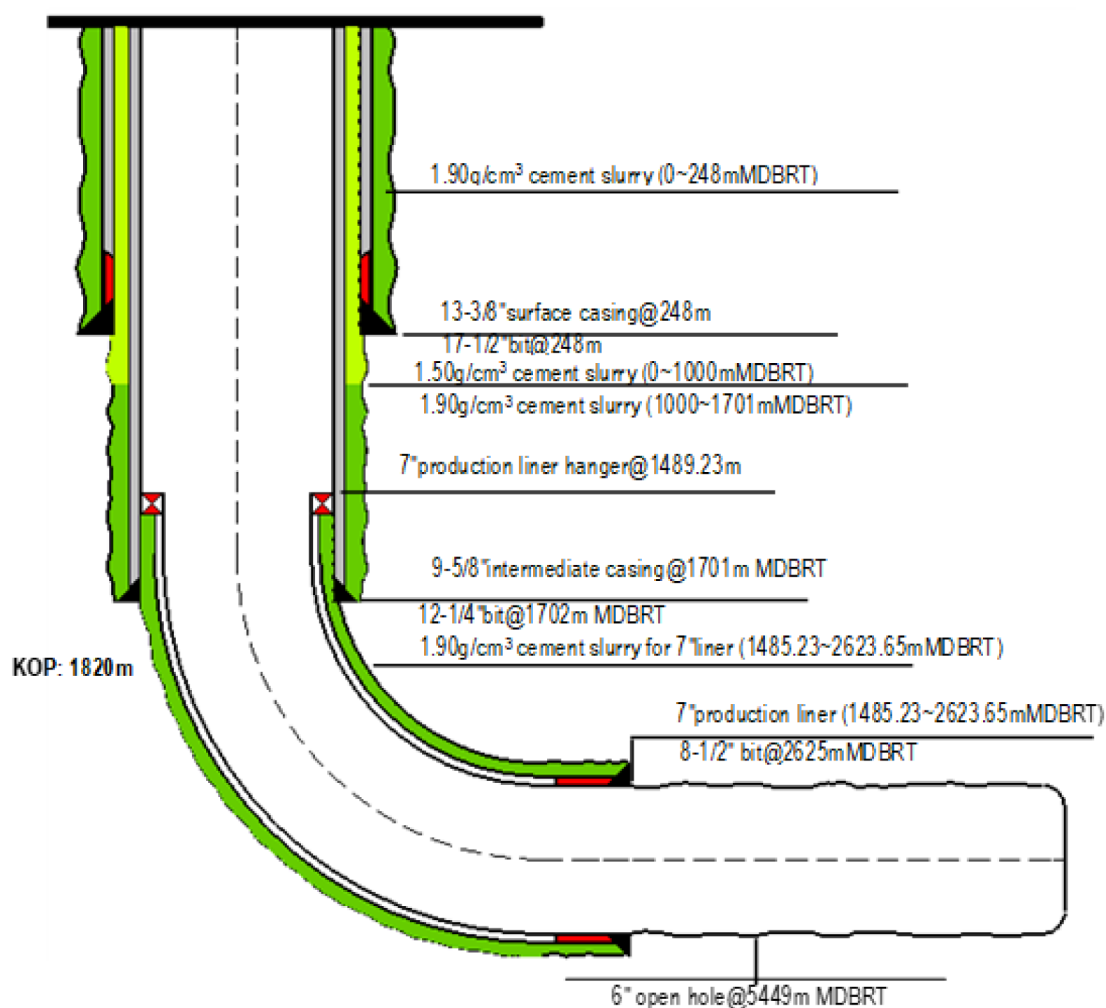


Figure 1—Actual well structure of EBST-26-2H. Due to the risk of potential BHA failure, the well was completed at MD 5449m, horizontal length 2826m. The planned horizontal section length of 3,000 meters was not achieved.

Table 1—Details of EBST-26-2H well structure

Section	Bit Size in	Casing Size in	Spud MD m	Casing MD Range m	Note
1 st	17-1/2	13-3/8	248	0-248	Surface casing protects shallow groundwater
2 nd	12-1/4	9-5/8	1702	0-1701	Isolate Lower Fars, prevent clay swelling and gypsum creep.
3 rd	8-1/2	7	2625	1489-2624	Production liner isolates formation before Target A
4 th	6	Openhole	5449	/	Openhole completion Reservoir protection

Drilling Challenges and Engineering Solutions

Despite thorough planning, drilling the 6-inch horizontal section of EBST-26-2H (2.826 km long) presented significant operational challenges. The primary issues were: high frictional drag and torque, hydraulic limitations in cuttings transport, and severe drillstring dynamics (stick-slip and vibrations) which led to RSS tool malfunctions. Additionally, secondary problems like hole cleaning, salt section swelling, and minor H₂S gas were encountered. This section details each challenge and the solutions implemented in the field.

High Friction and Torque in Extended Reach

The long 6" lateral created high frictional resistance and torque loads on the drillstring. Torque-and-drag modeling indicated an open-hole friction factor of ~0.20 while rotating and ~0.25 while sliding. Though moderate, over a 2.8 km horizontal these friction coefficients translated into substantial drag force and surface torque requirements. The use of small-diameter 3-1/2" S-135 drill pipe exacerbated friction effects due to its lower bending stiffness and greater contact area per unit length, increasing drag and risk of buckling.

Simulations predicted that rotating off-bottom (with no bit on rock) would require ~13.7 kN·m of surface torque just to overcome drag, while drilling ahead on bottom could raise total torque to ~14.5-15.8 kN·m. These values approach the mechanical limits of the drill pipe connections (the 3-1/2" tool joint make-up torque limit is ~17 kN·m). Indeed, the drilling program explicitly cautioned keeping surface torque below ~17 kN·m to avoid over-stressing the connections. Field data confirmed that torque and drag were the controlling factors in this extended-reach well (see Fig. 3 for the updated friction coefficient curve based on real-time drilling data): during the horizontal section, surface torque typically ranged from 8 to 14 kN·m and trended upward as measured depth increased. At several points (around 5,030-5,080 m MD), more than 100 kN of set-down weight was needed to slide the BHA, indicating near lock-up conditions and incipient helical buckling under compression. Weight on bit (WOB) also had to be limited (generally 20-50 kN) to prevent excessive torque spikes and tool overload.



Figure 2—A composite lubricant formula (combining liquid and solid components) is adopted to enhance the lubricity of drilling fluid and reduce torque and friction during long horizontal section drilling. (Source: Internal field data (EBS Petroleum,2023))

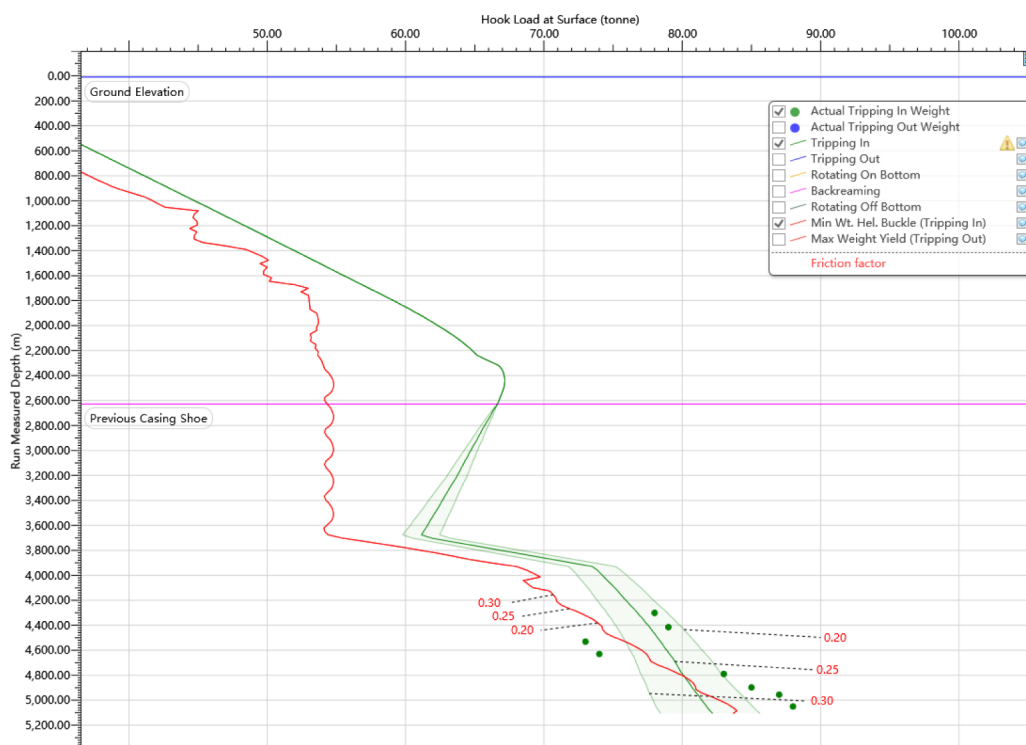


Figure 3—Update the friction coefficient of the BHA according the actual drilling data

Mitigation Strategies: A combination of design and operational measures enabled the well to reach TD without stuck pipe or over-torque failure:

Trajectory Optimization: The well path was planned and drilled with a gentle dogleg severity ($DLS \leq 4.5^\circ/30\text{ m}$) through the build and lateral, which yielded a smoother borehole with less side force on the pipe. Real-time surveys were closely watched to ensure the trajectory stayed within the planned glide path. This minimized unnecessary tortuosity that would increase friction.

Lubricated Mud System: The WBM was treated with high-performance lubricant additives (both liquid and solid types). Products such as EP-LUBE (extreme-pressure lubricant), graphite and plastic balls were added to reduce the coefficient of friction between drill pipe and wellbore. Daily lubricant concentrations were maintained as per program. These components form the composite lubricant formula illustrated in Fig. 2, which synergistically enhances mud lubricity to mitigate torque and friction during long horizontal section drilling.

Rig and String Modifications: Although stuck with 3-1/2" pipe for most of the well, the team strategically incorporated some 4" drill pipe toward the end of the lateral when it became available on site (during a bit trip). Replacing a portion of the string with 4" pipe in the upper part of the hole reduced overall string friction and allowed higher weight transfer to the bit without increasing torque (since 4" pipe has higher stiffness and critical buckling load). This hybrid string approach improved the ability to push the BHA in the final interval.

Friction-Reducing Practices: Continuous rotation was maintained whenever feasible (minimizing slide drilling) to reduce static friction. Frequent back-reaming and wiper trips were conducted - roughly every 300-500 m of advance - to ream out any tight spots and reduce cuttings beds that could increase drag. Standby time with the BHA stationary in the lateral was minimized; connections were made quickly and pipe movement (working the string up and down) was done before resuming drilling to alleviate any stickiness.

Through these combined measures, the effective friction factor was held around ~0.25-0.27 in the lateral, which while somewhat high, remained manageable. The drillstring remained free and movable throughout, and surface torque stayed just under critical thresholds. No twist-offs, over-torque incidents, or irretrievable stuck-pipe events occurred despite the record reach. The successful friction management in EBST-26-2H provides a template for similar slim-hole extended-reach wells.

Hydraulic Limitations and Hole Cleaning

Drilling a long 6" horizontal with 3-1/2" drill pipe also introduced major hydraulic challenges. The narrow annulus (only ~0.75" clearance in parts) and long flow path lead to high frictional pressure losses, limiting flow rates that the surface pumps could sustain. Pre-spud hydraulic modeling predicted that ~950 L/min (the desired flow for optimal hole cleaning) would generate standpipe pressures on the order of 4,200-4,300 psi - near the limit of the rig's mud pumps and surface equipment. In practice, to avoid overloading the system, the flow rate was kept between 830-850 L/min during the horizontal section, resulting in standpipe pressures of ~3,300-3,900 psi.

This flow rate was below the theoretical minimum for ideal cuttings transport in a 6" hole. Calculations indicated a critical cleaning flow of ~1,143 L/min in the horizontal to prevent significant cuttings bed deposition (assuming moderate cuttings size and density). Operating at ~75% of that ideal value posed a risk of solids accumulating in the low side of the wellbore. The team therefore employed multiple tactics to maximize hole cleaning efficiency under the flow-limited conditions:

Upgraded Pump and Surface Equipment: Prior to the lateral, the rig's mud pumps were fitted with larger liners (140 mm ceramic liners), effectively boosting their pressure rating to 35 MPa. High-pressure standpipe manifolds (rated 7,250 psi) were installed and successfully tested to 4,500 psi. These upgrades allowed the pumps to safely handle the ~3,800 psi pressures seen at 840 L/min, providing a small margin for occasional higher flow bursts when needed.

Optimized Fluid Rheology: The mud properties were tuned to carry cuttings efficiently even at moderate flow. Low-shear-rate viscosity (LSRV) was adjusted such that the 6 RPM Fann reading was ~10-12 (dimensionless units), roughly equal to the borehole diameter in inches - a rule of thumb for good suspension. This ensured the mud had sufficient gel strength to suspend cuttings during pauses. The yield point was maintained on the higher side (around 25-30 lbf/100ft²) for carrying capacity, but plastic viscosity was kept moderate to control friction pressure. This balance provided a highly carrying, yet pumpable, fluid.

High-Vis Sweeps and Periodic Trips: The crew pumped high-viscosity sweeps (viscosified gel pills) at regular intervals to help scour the low side of the hole and carry out cuttings. Additionally, short "cleaning trips" were scheduled approximately every 400-500 m in the lateral. The BHA would be pulled back ~200-300 m while back-reaming and then run forward to bottom, flushing the annulus. These trips, combined with back-reaming on connections, agitated and removed cuttings beds that may have formed.

Surface Solids Control: Both shakers and centrifuges were optimized to handle the increased solids load from the sweeps and reaming. Fine mesh shaker screens (API 200+) and continuous centrifuge operation ensured drilled solids were efficiently removed at surface so that the mud returning downhole was as clean as possible, maintaining its carrying capacity.

Even though the target flow rate could not be reached, these measures proved effective. No major pack-off incidents or excessive drag spikes (often indicative of cuttings buildup) were observed. Caliper logs and subsequent clean-out runs indicated the hole remained in gauge with minimal fill. The integration of hydraulic design (equipment and fluid tuning) with mechanical cleaning practices allowed safe drilling of the extended section without exceeding surface pressure limits.

Drillstring Dynamics and RSS Vibration Problems

The combination of high friction, extended reach, and a relatively slim drillstring led to notable drillstring dynamic issues, chiefly torsional oscillations (stick-slip) and associated vibrations. Surface torque while drilling exhibited cyclic fluctuations: e.g., oscillating between ~8 kN·m and 14 kN·m in a periodic fashion (Fig. 4 shows the downhole stick-slip data recorded by the RSS system), indicating severe torsional oscillations. These wide swings are characteristic of stick-slip behavior, where the bit momentarily stalls ("sticks") causing torsional wind-up in the drillstring, then suddenly breaks free ("slips"), unleashing the stored torque. Downhole sensors (in the BHA's vibration module) registered stick-slip severity index (SSI) values exceeding 1.5 at times (on a dimensionless scale where >1.0 indicates severe torsional oscillation), and lateral shock readings above 2.0 (in units of g or a normalized scale). Such dynamic dysfunction not only reduces drilling rate (as energy is wasted in vibrations) but can also damage downhole tools and bits.

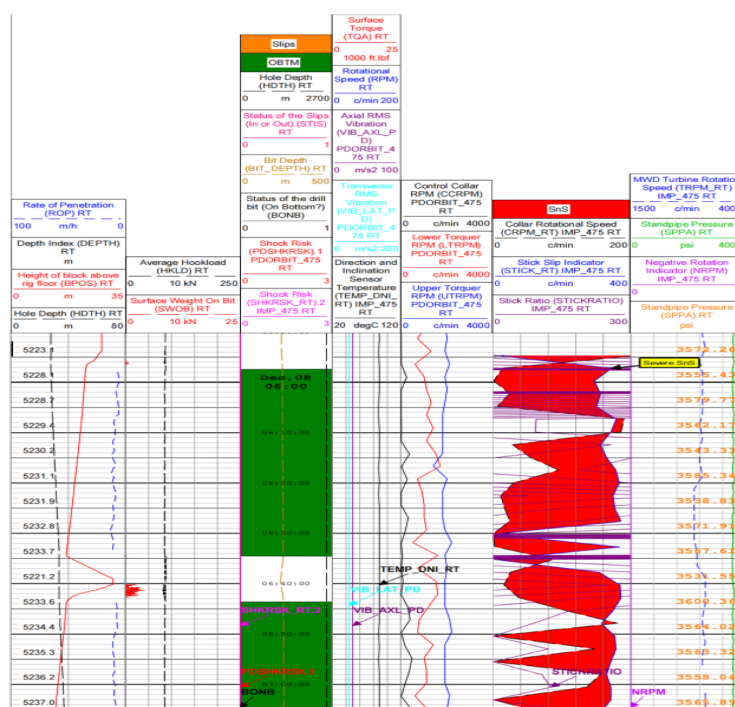


Figure 4—The actual stick and slip data recorded by the RSS system

The use of 3-1/2" drill pipe likely exacerbated the issue - its lower polar moment of inertia provides less inherent damping of torsional vibrations. Additionally, the long horizontal section acted as a massive compliant spring: the farther the bit drilled out, the more stretch and wind-up in the system, amplifying the stick-slip tendency. These vibrations were not merely theoretical; they had a direct operational consequence: the rotary steerable system tool experienced multiple malfunctions attributable to the harsh dynamic environment.

Mitigation and Adjustments: After the first RSS failure due to vibration (at ~4,616 m MD), the drilling team adjusted parameters and procedures to dampen stick-slip:

Higher Rotary Speed: Paradoxically, increasing rotary RPM can mitigate stick-slip by reducing the time for stick events to grow. The rotary speed was raised from ~90 RPM to about 110-120 RPM for the remainder of the lateral. The higher rotary speed shortened the stick-slip oscillation period and tended to keep the bit in continuous motion, which reduced the amplitude of torque cycles.

Controlled WOB: Weight on bit was moderated in the 20-50 kN range, even when the formation might have drilled faster with more weight. This was to avoid excessive bit force that could dig the bit in and trigger stick episodes. By keeping WOB moderate (and occasionally backing off weight when torque spikes were observed), the bit was less prone to sticking hard in the rock.

Drilling Dysfunction Monitoring: Real-time downhole vibration data (when available from the RSS/LWD tools) and surface indicators (torque variance, rate of penetration fluctuations) were closely monitored. Anytime stick-slip indicators worsened, the driller would make immediate changes - e.g., further increase RPM, reduce WOB, pump a lubricant sweep, or perform a short off-bottom rotation to relax the string.

Enhanced Hole Conditioning: As described earlier, frequent short trips and aggressive hole cleaning were done. This also helped vibration: a clean, gauge hole reduces the chance of the BHA getting intermittently stuck or experiencing differential sticking forces that can aggravate torsional oscillations. After the initial RSS incident, wiper trips were done even more frequently (every ~300 m) and back-reaming was employed whenever coming off bottom.

With these adjustments, the drillstring dynamics improved. Stick-slip was not eliminated, but its severity was reduced to manageable levels (estimated SSI generally ≤ 0.5 -1.0 after changes). Importantly, the remaining ~800 m of the lateral was drilled without further major tool failures attributed to vibration. The ability to "tune" the drilling parameter combination on the fly - essentially finding a sweet spot of RPM, WOB, and flow - was crucial to navigate the last portion of the well safely.

Secondary Challenges: Hole Cleaning, Salt Swelling, and Sour Gas

In addition to the primary mechanical and hydraulic issues above, several secondary challenges were encountered and managed during the operations:

Hole Cleaning in Horizontal: Due to pump capacity limit, the flow rate was sub-optimal for cuttings transport. This risked cuttings bed buildup, especially in the ~90° inclination section. The mitigations implemented (high-vis sweeps, frequent short trips, etc.) proved effective. Measurements while drilling and after drilling indicated that cuttings bed height remained minimal (estimated <5 cm or ~2 inches on the low side) and no incidents of stuck pipe occurred. The success in hole cleaning despite flow limitations underscored the importance of mechanical removal (via back-reaming) to complement hydraulic transport.

Salt/Gypsum Stringers: In the upper hole, the Lower Fars formation (~950-1,375 m MD) contains thick gypsum/anhydrite layers. These formations can absorb water and swell, or dissolve and precipitate, potentially causing pack-off or pipe sticking. While drilling this interval, the mud system was treated with inhibitors and the pH maintained >9.5 with caustic soda to reduce any clay swelling. Solids control was closely watched to avoid contaminating the mud with fine reactive salt particles. The well experienced minor tight hole symptoms coming out of Lower Fars, but increased flow and an extra reaming run alleviated the drag. No significant issues were encountered from the salt section, and it was cased off by the 9 5/8" intermediate casing.

H₂S Occurrence: While offset data warned of high H₂S in deeper formations, EBST-26-2H encountered only very low levels of H₂S (a few tens of ppm maximum in mud gas) while drilling the Tanuma. These traces were quickly captured by the scavenger chemicals in the mud. Continuous monitoring verified that toxic gas remained well below alarm thresholds. Nonetheless, the stringent H₂S safety protocols were maintained throughout as a precaution. The operation had zero H₂S-related incidents, validating the proactive safety approach.

By the time the well reached total depth (TD) at 5,449.6 m, the integrated application of these engineering solutions allowed the project objectives to be met without major non-productive time (NPT). The 6" open hole was conditioned successfully for completion - for example, a 2" coiled tubing string was later run to TD for an acid stimulation with no issues, indicating a smooth, clean wellbore. The field trial of this long horizontal well thus demonstrated that even under equipment and environmental constraints, careful planning and adaptive execution can deliver record-length laterals.

Rotary Steerable System (RSS) Malfunctions and Remedial Strategies

Drilling the extended horizontal section with a rotary steerable system was critical for trajectory control, but the RSS tool experienced multiple malfunctions due to the challenging downhole conditions. EBST-26-2H utilized a point-the-bit RSS for 6" slimhole drilling, chosen for its aggressive build capability and slim diameter. Despite this advanced tool, the long reach and vibration environment led to three discrete failure events during the run. This section describes the nature of each RSS failure, the immediate remediation actions taken, and the lessons learned for future operations.

Nature and Timing of Failures

Failure Event 1 (Dec 1, 2023, at ~4,616 m MD): Shortly after the RSS BHA entered the horizontal section, the tool failed to respond to steering commands. Surface telemetry indicated loss of communication with the RSS's steering unit. Diagnostic tests showed the power-drive (PD) module was not generating the necessary power or signals for tool actuation. Based on collaboration with the service provider, the likely cause was a blockage of the RSS's internal turbine and generator system by debris. Fine cuttings or solids had probably accumulated around the tool's intake ports, impeding the mud-driven turbine that powers the steering electronics. This effectively shut down the RSS's ability to change bias or hold inclination. Such an occurrence is consistent with debris-related RSS failures reported in field drilling experience [6]. The loss of power/communication in this event prevented any steering until remedial action was taken.

Failure Event 2 (Dec 3, ~4,778-4,922 m MD): While drilling in a Hold Inclination Angle (HIA) mode, the RSS gradually lost directional control over a span of approximately 150 m. The inclination, which was supposed to hold ~89.5°, drifted down to ~88.4° despite no change in surface commands. At the same time, downhole vibration sensors recorded a stick-slip severity >1.5 and shocks >2.0. This suggests that severe torsional oscillations were occurring, likely disrupting the RSS's control loops or sensor readings [7]. High-frequency torsional vibrations can momentarily make the tool unresponsive or cause it to enter a safe mode. In this incident, the RSS did not outright fail mechanically, but it effectively ceased adjusting the trajectory (stuck in HIA mode) for a period, leading to the observed drop in angle. The timing coincided with drilling through a harder streak that caused more stick-slip. The lack of immediate failure meant drilling continued, but the trajectory deviation alerted the team to a dysfunction in progress.

Failure Event 3 (Dec 7, ~5,155-5,221 m MD): Later in the lateral, the RSS again became unresponsive - this time it was unable to transition from hold mode to a planned build section. The inclination fell from ~89.4° to ~88.1° (losing angle) when a build command was issued and the tool failed to execute it. This occurred after a short trip had been done without a top-of-BHA debris filter, potentially allowing new debris to settle in the tool. Post-run analysis indicated that fine cuttings had likely entered the RSS's control valve mechanism during the trip, similar to the first event, and jammed the steering ribs in a fixed

position. Essentially, the RSS was "stuck" in the last commanded state and would not respond to further mode changes. This third event occurred near the end of the section, and the decision was made to pull out and replace the RSS tool to complete the well.

Notably, all three incidents happened at extended depths beyond ~4.6 km, where the drilling dynamics and debris loads were most severe. None of these failures caused a total loss of drilling progress (the well still reached TD), but they resulted in trajectory deviations and required time for troubleshooting and corrective action. The first two were resolved without pulling the BHA, whereas the third necessitated a tool change. These experiences underscored that in ultra-long horizontals, RSS tools may face unique reliability issues mainly due to (a) debris accumulation in the tool, (b) extreme vibration affecting tool electronics and sensors, and (c) weak signal telemetry at long distance.

Remedial Actions and Functional Recovery

The engineering team employed a consistent set of remediation strategies to address the RSS malfunctions in real time and restore directional control:

High-Rate Circulation Flush: Upon suspecting a debris-related RSS stall (as in Failure 1 and 3), the well was circulated at maximum safe rates (~830-850 L/min) for an extended period (1-2 hours) with the bit off bottom. This high-flow flushing was intended to sweep away any cuttings or fines around the BHA and inside the tool's ports. In both the first and third events, this action was taken immediately. In the first event, it was done in place, and in the third, it was done prior to pulling the tool. The aggressive circulation likely dislodged some obstructions, as evidenced by improved flow and pump pressure behavior afterward.

Short Tripping the BHA: During the first RSS failure, the crew executed a short trip: pulling the drill string back ~300-400 m and then running back to bottom. This maneuver (essentially reaming the same interval twice) physically dragged the BHA through the problematic zone and helped clear accumulated cuttings. The process was done while pumping at high rates and rotating to maximize agitation. Impressively, this was accomplished without sticking the pipe. After this short trip and return to the original depth, a surface test indicated the RSS had regained functionality (steering commands were acknowledged). For the second event, a similar approach was taken preemptively when stick-slip was noticed - the string was worked and partially tripped to reduce torsional stress and then drilling resumed, which helped avoid a persistent failure.

Lubricant Slurry Pill: In conjunction with the flushing for the second event, a specialized lubricant pill was pumped. This pill contained fine graphite and plastic beads designed to reduce friction in the BHA and stabilize flow. The idea was to both lubricate any moving parts in the RSS and also to sweep through any tight annulus sections with a low-friction fluid, possibly helping the RSS's internal components move freely again [7]. While it's hard to directly quantify its effect, the lubricant pill likely contributed to the RSS recovering from the second event by easing the torque load and damping vibrations.

Parameter Tuning: Drilling parameters were adjusted in response to RSS issues. Specifically, rotary RPM was increased (~110 RPM) to reduce low-frequency stick-slip, and WOB was reduced by about 15-20% from prior levels to avoid large torque oscillations. These changes were instituted immediately after detecting control loss (especially in Event 2) and are credited with preventing a complete failure in that case [8]. Essentially, by "backing off" the aggressiveness of drilling (less WOB) and spinning faster, the dysfunction did not escalate.

BHA Positioning for Signal: There were instances during the first and second events where surface commands to the RSS were not acknowledged. The crew responded by partially picking up the drill string to shallower depth (by a few hundred meters) to improve the telemetry signal path, then re-sending commands. This technique exploited the fact that signal attenuation is worse at deeper depths; pulling the tool up reduces the transmission distance and can bypass any downhole signal interference. In at least one case, raising the BHA allowed a previously unacknowledged command to go through successfully, restoring control.

Progressive Pipe Replacement: After the second RSS incident, some 4" drill pipe stands were added during a trip. This reduced the overall torsional compliance and may have contributed to fewer problems

afterward. It also lowered the circulating pressure slightly, which could have helped keep flow up when it was most needed.

Using the above measures, the team managed to recover RSS functionality without pulling out of hole for the first two incidents. The RSS resumed normal operation and remained online for over 150 circulating hours after the first failure, and about 80 hours after the second, before the final failure occurred. Only the third event ultimately necessitated pulling the BHA and changing the RSS tool. These remedial actions demonstrated that even when an RSS appears to fail, it may be possible to avoid a costly trip by applying creative circulation and mechanical techniques to clear problems downhole.

Functional Testing and Confirmation

After each intervention that was intended to fix the RSS, the team performed a series of functional tests to ensure the tool was indeed working properly before resuming full drilling:

Micro-Command Tests: The first check was to send small steering commands (e.g. a $\pm 0.5^\circ$ inclination change or a minor toolface offset) and observe whether the tool responded. If the RSS was communicating and able to actuate, these micro-commands would result in a measurable change in tool readings (inclination or azimuth trend). For example, after the first short trip flush, a slight toolface adjustment was commanded and confirmed at surface via the downhole telemetry.

Pilot Drilling: Rather than immediately continuing the section at full bore, the crew drilled a short distance (5-10 m) as a pilot test. During this pilot, they closely watched the trend of inclination and toolface. In one case (after the second event), they drilled ~ 30 m with minimal corrections to verify the RSS could hold angle again. This pilot drilling served to demonstrate that the well trajectory was once more under control (in Event 2, inclination stabilized around $89.5^\circ \pm 0.1^\circ$ after intervention, confirming the RSS was back steering properly).

Steering Hold Verification: The tool was left in hold inclination mode over multiple stands to see if it would maintain the angle or if any drift occurred. For instance, after the remedial measures on Dec 3, the inclination held steady ($\sim 89.5^\circ$) over the next 90 m without dropping, whereas before it had been falling. Additionally, torque fluctuations were much reduced (to < 1 kN·m swings), indicating the stick-slip had been tamed. Together, these signs gave confidence that the RSS dysfunction had been resolved (at least temporarily).

Only once these tests were satisfactorily passed did the team ramp back up to normal drilling progress. This cautious approach ensured that any remaining RSS issues would be caught early rather than drilling a long interval unknowingly with a non-functioning tool.

Engineering Insights and Recommendations

The repeated RSS malfunctions in EBST-26-2H, while challenging, provided valuable lessons for future extended-reach drilling with RSS tools:

Debris Management is Crucial: Two of the three RSS failures were linked to debris plugging the tool's internals. A clear takeaway is the need for drill pipe filters and better downhole filtration when tripping or reaming in long laterals. Installing a simple coarse screen sub at the top of the BHA could prevent larger cuttings from entering the RSS on trips. In hindsight, the absence of a filter during the short trip likely contributed to the third failure. Going forward, running a drillstring filter any time an RSS is in hole for extended horizontal drilling is recommended.

Long-Distance Telemetry Limitations: Extended-reach wells push the limits of mud pulse telemetry. The partial command losses experienced suggest that beyond $\sim 4,500$ m, signal strength can degrade. Techniques such as temporarily pulling the BHA up to resend commands (as was done) or using wired drill pipe or other amplification technologies should be considered. If available, a wired telemetry system would likely have prevented some delay in detecting and correcting the issues [9]. Absent that, planning for slower communication and possibly repeating commands is necessary at extreme depths.

Vibration Mitigation Protects Tools: There was a direct correlation between high stick-slip and RSS control issues. The measures to reduce vibration (increased RPM, lubricant sweeps, etc.) not only improved drilling performance but also were instrumental in keeping the RSS functional. This highlights that controlling drilling dysfunctions like stick-slip is not just about ROP, but also about tool reliability. Future wells should include real-time vibration monitoring and set threshold alarms (e.g. if downhole SSI > 1.0 for a sustained period, take corrective action) to preempt tool damage.

Standardized Contingency Workflow: Perhaps the most important outcome was the development of a stepwise contingency protocol for RSS/power drive failures. By codifying the actions - high flow circulation, working the pipe, parameter changes, etc. - the team created a playbook for handling similar events. This proved effective when a second and third failure occurred, as the driller and engineers knew immediately how to respond. Establishing such a "recipe" for intervention has been adopted for future ERD wells in the field.

In summary, the RSS challenges in EBST-26-2H were overcome through adaptive operations and yielded a set of best practices. Despite multiple scares, none of the incidents led to significant non-productive time, and the well was steered to target successfully. The engineering ingenuity applied (from stick-slip control to on-the-fly tool recovery) is a testament to robust contingency planning and real-time problem-solving in complex drilling environments.

Contingency Protocols for Downhole Dysfunction

Extended-reach wells like EBST-26-2H demand not only initial design planning but also prepared contingency measures to address unplanned downhole problems. Based on the experiences in this well - particularly the RSS malfunctions and drilling dysfunctions - the team developed a standardized contingency workflow to handle downhole tool failures or signal losses. This protocol provides a step-by-step guide for diagnosis and remediation, ensuring a rapid and effective response that minimizes downtime. Key elements of this contingency plan and the broader lessons learned are outlined below.

Standard Workflow for RSS Signal Loss and Recovery

If a loss of RSS responsiveness or communication is detected (through telemetry or observed trajectory deviation), the following staged workflow is implemented:

1. **Diagnosis.** Confirm dysfunction via surface logs (no response to downhole commands, or unexpected trajectory drift). Check surface data (torque, hook load, vibration indices) for signs of stick-slip or excessive drag. Verify surface equipment is functioning to rule out surface telemetry issues.
2. **Initial Remediation.** Pump at maximum safe circulation rate (~800–850 L/min) to flush debris. Work the drillstring gently (pick up/slack off, rotate slowly) while maintaining circulation. Pump a lubricant pill (1–2 m³ with high lubricant concentration) to ease internal friction.
3. **Enhanced Response.** If still unresponsive, perform a short trip: pull out 300–400 m while rotating and circulating, then re-run to bottom. Increase rotary RPM (~110) to dampen stick-slip. Reduce WOB and pause drilling until tool status is stable.
4. **Verification & Recovery.** Resend basic commands (small inclination or toolface adjustments). If telemetry is weak, pick up the drillstring slightly to improve signal strength. Verify tool response by observing immediate downhole feedback.
5. **Post-Recovery Monitoring.** Resume drilling cautiously, closely monitoring parameters and vibration data. Avoid frequent steering commands until stable operation is confirmed. Prepare for recurrence and plan for tool replacement if instability persists.

By following this structured workflow, the team on EBST-26-2H was able to handle the RSS malfunctions systematically. The approach emphasizes first addressing any drilling-induced causes (vibration, debris)

and then verifying tool responsiveness before proceeding. This prevented knee-jerk pulling of the BHA - saving considerable rig time - and ensured safety by stabilizing the situation before continuing.

Lessons Learned and Best Practices

Implementing the above contingency measures and reflecting on the entire operation led to a number of key lessons and best practices for future extended-reach and cluster drilling projects in the field:

Prevent Debris Ingress: A small investment in preventive tools, like drill pipe filters or dual float valves with screens, can significantly reduce the risk of RSS and BHA component failure due to solids. All future RSS runs will include a top-of-BHA filter sub, especially during any short trips or backreaming operations, to catch cuttings before they can settle inside tools.

Signal Loss vs. Tool Failure: It became evident that many "tool failures" are actually signal or power issues rather than physical breaks. In EBST-26-2H, none of the RSS incidents were due to mechanical breakage; rather, they were due to power loss or control interruptions. This suggests that when an RSS goes unresponsive, operators should not immediately assume the tool is broken. Quick remediation (flushing, etc.) can often restore function, as we saw. This nuance will be communicated in training: treat the first sign of RSS issues as potentially recoverable, not as immediate non-productive time (NPT).

Stick-Slip Control Pays Off: The benefits of reducing stick-slip went beyond improving ROP - it also preserved equipment health. The link between high stick-slip and RSS trouble was clearly demonstrated. Therefore, the practice of monitoring vibration indices in real time and actively managing parameters to keep stick-slip low will be standard. Use of surface vibration dampeners, torsional oscillation tools in the string, or even real-time advisory systems to adjust RPM may be explored to further mitigate stick-slip in future wells[7][8].

Standardized Contingency Plans: Having a written contingency plan (like the workflow above) significantly improved team response and confidence. By codifying the steps and assigning responsibilities (e.g., mud engineer prepares lubricant pill, directional driller handles re-send of commands, etc.), the team functioned efficiently under pressure. This approach is now part of the drilling program for upcoming wells: the program document will include a "Contingency and Troubleshooting" section for common issues (losses, stuck pipe, RSS faults, etc.) so that rig crews have guidance at hand.

Overall, these strategies showcase effective real-time engineering under challenging conditions. Through preparedness and adaptive techniques, the EBST-26-2H team was able to continue operations with minimal disruption despite complex downhole dysfunctions. The knowledge gained has been fed back into future well planning, meaning subsequent extended-reach wells in the block can be executed with greater confidence and improved procedures.

Multi-Objective Drilling Parameter Optimization Using NSGA-II

Data-Driven Modeling for ROP and Stick-Slip

Well Data and Features: The EBST-26-2H well (an extended-reach horizontal well) provided a comprehensive field dataset for our study. Key drilling parameters - WOB, (kN), rotary speed (RPM), and pump flow rate (Q, L/min) - were used as input features, along with other operational measurements (standpipe pressure, torque, etc.). These controllable inputs strongly influence drilling performance and vibrations. The data was quality-checked and split into training (70%) and validation (30%) sets to develop predictive models. Prior to modeling, features were normalized and outliers (e.g. transient values during pipe connections) were removed to improve model generalization ability

Modeling Workflow and Data QA/QC

The end-to-end workflow used to build the ROP and stick-slip predictive models is summarized as follows:

1. Data collection & consolidation: Rig sensors, surface & downhole logs, mud reports; align depth/time.
2. QA/QC & signal conditioning: Remove connections/off-bottom; despike; handle gaps; synchronize channels; cap outliers; normalize features.
3. Feature engineering: Construct controllable inputs [WOB, RPM, Q] and supportive features (SPP, torque, ECD proxies); compute SSI target from RPM fluctuations if needed.
4. Leakage-free dataset split: Block-wise 70/30 split by time/depth; ensure no overlap of transients.
5. Model training: Random Forest (RF) predict ROP; Gradient Boosting Regressor (GBR) predict stick-slip; hyperparameter tuning.
6. Validation & k-fold CV: Report R^2 /MAE; residual checks; parity plots; stability across folds.
7. NSGA-II optimization Objectives: maximize ROP, minimize SSI; bounds on WOB/RPM/Q; produce Pareto-optimal recommendations.

QA/QC highlights. removed off-bottom and connection intervals, spurious spikes, and duplicated records; synchronized depth- and time-based channels; imputed short gaps and dropped long gaps; normalized features and capped extreme outliers; and verified physical bounds (e.g., torque, SPP, and flow) prior to modeling.

Leakage-free splits. To avoid temporal leakage in time-adjacent samples, we split data by contiguous blocks (70/30) and verified that transients used for training do not appear in validation. Generalization using k-fold cross-validation is further assessed.

Metrics and acceptance. Following QA/QC, the RF achieved high agreement ($R^2 \approx 0.85\text{--}0.90$; $\text{MAE} \approx 0.5\text{--}1.0$ m/h) and the stick-slip GBR achieved $R^2 \approx 0.80$ on validation, consistent with ensemble models' ability to capture nonlinearities (see Fig. 5). These models are then coupled with NSGA-II for multi-objective optimization.

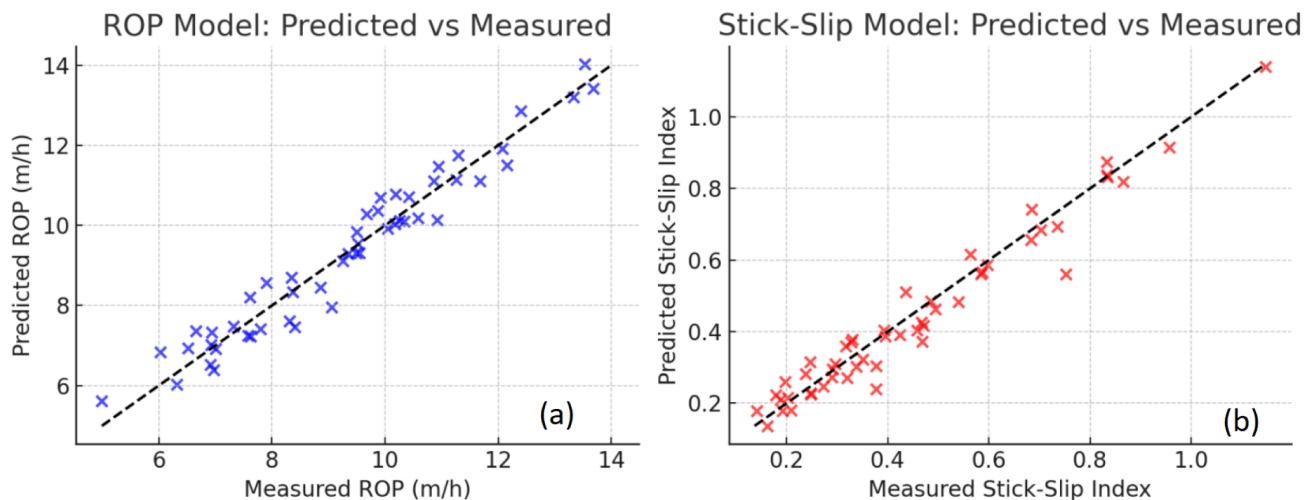


Figure 5—Model performance on validation data. (a) Random Forest ROP model predictions vs. measured ROP (m/h), and (b) Gradient Boosting model predictions vs. measured stick-slip index.

ML Models for ROP and Vibration

A RF regression model (an industry-standard open-source ensemble learning algorithm) was trained to predict the ROP, and a GBR model (another widely used open-source ensemble method) was trained to predict a stick-slip severity index. These ensemble learning methods were chosen for their ability to capture nonlinear relationships and interactions among drilling parameters. The RF model for ROP used decision-

tree ensembles to average predictions, which improves robustness against overfitting and provides feature importance measures. The GBR model for stick-slip builds trees sequentially to minimize error, capturing subtle effects on torsional vibration. Both models took the same input vector $x = [\text{WOB}, \text{RPM}, Q, \dots]$ and produced their respective targets.

Model Training and Accuracy: The ROP training achieved a high coefficient of determination (e.g., $R^2 \approx 0.90$ for training and 0.85 for validation), indicating the RF model could explain approximately 85-90% of ROP variance. The Mean Absolute Error (MAE) on validation was 0.5-1.0 m/h, which is acceptable given the typical field ROP range of 5-15 m/h. The stick-slip index model showed slightly lower accuracy (validation $R^2 \approx 0.80$), reflecting the stochastic nature of downhole vibration.

Fig. 5 presents representative parity plots comparing model predictions with measured values for both targets. The data points cluster tightly around the 45° line with no systematic bias, demonstrating that the models effectively capture underlying trends (correlation ≈ 0.92 for ROP and ≈ 0.89 for stick-slip). The ensemble models outperformed single regressors such as individual decision trees, consistent with findings in existing literature.

For further validation of generalization, k-fold cross-validation was conducted. Results showed consistently low errors, and feature influences were physically interpretable. For example, the RF model's permutation importance ranked flow rate and WOB as the most influential factors for ROP, aligning with domain expectations.

Points near the 45° line indicate good agreement, with R^2 values around 0.85-0.90 for both models. Dashed line is $y=x$ for reference. **Stick-Slip Index Definition:** It is important to define the stick-slip severity metric used as the optimization target. We adopt a stick-slip index (SSI) defined as the ratio of rotary speed fluctuation to average rotary speed:

$$\text{SSI} = \frac{\Omega_{\max} - \Omega_{\min}}{2\Omega_{\text{avg}}} \quad \text{Eqs.(1)}$$

where $\Omega_{\max}$, $\Omega_{\min}$, and Ω_{avg} represent the maximum, minimum, and average downhole rotational speeds (or RPM) over a short time window (e.g., 16-second intervals). This dimensionless index approaches 0 during steady rotation and increases toward 1.0+ as stick-slip oscillations intensify.

The SSI can be derived from surface and downhole sensor data: values above 0.5 typically indicate significant torsional oscillation, while those below 0.2 represent low vibration levels. The GBR model was trained on SSI values calculated from high-frequency downhole measurements (or inferred from surface torque/RPM oscillations), enabling it to predict vibration severity based on drilling parameters.

Multi-Objective Optimization Problem Formulation

With the predictive models established, we formulated a multi-objective optimization framework to maximize drilling rate while minimizing vibrational risk. The decision vector is $x = [\text{WOB}, \text{RPM}, Q]$ - the controllable drilling parameters to be optimized. Two competing objective functions were defined based on the machine learning models:

Maximize ROP: $f_1(x) = \text{ROP}_{\text{rf}}(x)$, representing the predicted rate of penetration (m/h) from the Random Forest model

Minimize Vibration: $f_2(x) = \text{SSI}_{\text{gbr}}(x)$, representing the predicted stick-slip index (dimensionless) from the Gradient Boosting model

In summary, the optimal solutions

$$x^* = \arg \max_x f_1(x) = \arg \min_x f_2(x) \quad \text{Eqs.(2)}$$

and $\arg \min f_2(x)$ conflict with each other - higher WOB and RPM increase ROP but tend to elevate vibration severity - meaning no direct single-objective optimum exists. Instead, we seek the Pareto-optimal set of

solutions where improving one objective would necessarily worsen the other. Mathematically, the problem is formulated as below, while other operational constraints are also applicable

$$\text{Maximize } f_1(x) = \text{ROP}_{\text{rf}}(x) \quad \text{Eqs.(3)}$$

$$\text{Maximize } f_2(x) = \text{SSI}_{\text{gbr}}(x) \quad \text{Eqs.(4)}$$

Subject to:

$$\text{WOB}_{\min} \leq \text{WOB} \leq \text{WOB}_{\max} \quad \text{Eqs.(5)}$$

$$\text{RPM}_{\min} \leq \text{RPM} \leq \text{RPM}_{\max} \quad \text{Eqs.(6)}$$

$$Q_{\min} \leq Q \leq Q_{\max} \quad \text{Eqs.(7)}$$

Parameter Bounds and Constraints: Bounds were set based on field operating limits and safety considerations. For EBST-26-2H, the ranges used were: WOB from 20 kN to 100 kN (approximately 2-10 tons weight-on-bit), RPM from 60 to 150 (based on top-drive capabilities and bit design), and pump flow Q from 500 to 1000 L/min (limited by pump horsepower and hole cleaning requirements). These ranges encompass the normal drilling window - for example, the minimum WOB ensures constant bit contact, while the maximum WOB prevents exceeding bit/tool capacity or inducing stick-slip oscillations from excessive bit pressure.

In addition to these simple bounds, we imposed a constraint that the predicted stick-slip index must remain below 1.0 (to avoid severely damaging oscillations where the bit fully sticks). In practice, this constraint was consistently satisfied by solutions on our Pareto front (all had $\text{SSI} \leq 0.5$), meaning the multi-objective optimization inherently kept vibration within a safe range. Other constraints included maintaining standpipe pressure at a level that ensures downhole pressure stays below the expected fracture gradient, as well as keeping bit hydraulic horsepower within operational limits – these, however, were indirectly managed by capping the flow rate and WOB. Mud weight and bit type remained fixed during optimization, so their effects (e.g., on MSE or bit wear) fall outside the scope of this model.

Objective Function Equations: The RF and GBR surrogate models replace costly physical simulations or field trials, enabling rapid evaluation of f_1 and f_2 for any candidate parameter set x . For example, the RF model might produce an explicit mapping like $f_1(\text{WOB}, \text{RPM}, Q) \approx 15.0 \times [1 - \exp(-0.05 \times \text{WOB})] \times [1 - \exp(-0.03 \times \text{RPM})] \times (Q/1000)$ (a learned functional form) that shows diminishing ROP returns at high WOB and RPM. Similarly, the GBR model might capture a relationship approximately like $f_2 \propto \text{WOB}/\text{RPM}$, indicating that stick-slip worsens with higher weight and lower rotary speed.

Fig. 6 illustrates these relationships: the left panel shows predicted ROP as a function of WOB and RPM, while the right panel shows the predicted stick-slip index across the same parameter space (with flow rate fixed). ROP increases toward the top-right (high WOB, high RPM), whereas the vibration index increases toward the bottom-right (high WOB, low RPM). This inverse trend between ROP and stick-slip confirms the need for multi-objective trade-off analysis.

Notably, flow rate Q had a positive but secondary effect on ROP (higher Q slightly improves hole cleaning and thus ROP) and a negligible effect on stick-slip (except indirectly through bit hydraulics and differential pressure). Consequently, the optimization consistently pushes Q to its upper bound to maximize ROP, with the primary trade-off occurring between WOB and RPM. We therefore focus on the (WOB, RPM) plane when visualizing solutions

On the left of Fig.6, predicted ROP (m/h) increases with both WOB and RPM, exhibiting diminishing returns at high values (evident in flattening contours). On the right of Fig.6, predicted stick-slip severity index rises with WOB but decreases with RPM (higher RPM helps mitigate torsional oscillations). These conflicting objectives form a Pareto frontier, where improving ROP necessarily comes at the expense of increased vibration.

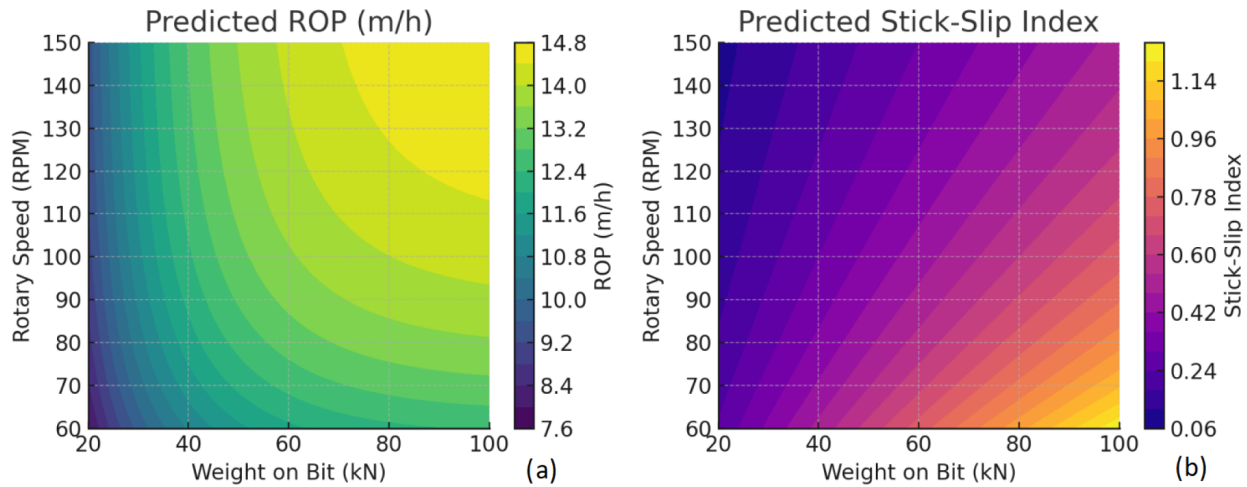


Figure 6—Modeled objective landscapes as functions of controllable parameters (flow rate fixed at 800 L/min):

NSGA-II Optimization Algorithm Implementation

To solve the above multi-objective problem, we employed the Non-dominated Sorting Genetic Algorithm II (NSGA-II), an open-source, widely adopted multi-objective evolutionary algorithm. As a fast elitist evolutionary algorithm, NSGA-II can find a set of Pareto-optimal solutions in a single run. It adopts a population-based approach inspired by natural selection, evolving candidate combinations of drilling parameters toward the Pareto front that balances maximum ROP and minimum stick-slip. NSGA-II was chosen for its proven effectiveness in similar engineering optimization scenarios and its ability to maintain diversity among multiple solutions (via the crowding-distance mechanism).

Algorithm Parameters: Table 2 summarizes the NSGA-II settings. Each solution $x = [WOB, RPM, Q]$ was encoded as a real-valued chromosome. The initial population (size $N=100$) was generated using Latin Hypercube sampling within the allowed parameter ranges to ensure broad coverage of the search space. The algorithm ran for 200 generations, which was sufficient for convergence (the Pareto front stabilized after approximately 150 generations).

Table 2—NSGA-II parameters for drilling optimization

Parameter	Value	Description
Population size, N	100	Number of candidate solutions per generation.
Generations	200	Stopping criterion (or until Pareto front converges).
Crossover type (probability)	Simulated binary (0.9)	Recombination operator to create offspring (90% chance).
Mutation type (probability)	Polynomial (0.1 per gene)	Small random perturbations for exploration (10% chance per variable).
Selection	Binary tournament	Selects parents based on non-domination rank and crowding distance.
Elitism	Yes (NSGA-II elite archive)	Ensures Pareto-optimal solutions are retained each generation.

For real-coded genetic operations, we used simulated binary crossover (SBX) and polynomial mutation: the crossover probability was 0.9 (meaning 90% of offspring were produced by mating), and the mutation probability was 0.1 per gene. Binary tournament selection was adopted, where individuals are ranked first by non-domination level and then by crowding distance (to promote diversity). Elitism is inherent in the algorithm: at each generation, parent and offspring populations are merged, and the top N non-dominated

solutions are carried forward. This ensures the Pareto front never degrades - a key feature of NSGA-II's elitist strategy.

Optimization Results and Pareto Front Analysis

After running NSGA-II, we obtained a set of Pareto-optimal solutions that span the trade-off between ROP and stick-slip severity. Fig.7 plots the Pareto front in the objective space (predicted ROP vs. predicted stick-slip index), along with dominated candidate solutions for comparison. The Pareto front (red points) is concave, reflecting a classic diminishing returns relationship: increasing ROP further leads to an increasingly steep rise in vibration severity. Red circles represent non-dominated solutions forming the Pareto optimal curve, while grey crosses denote other (dominated) candidates explored during optimization. The front clearly illustrates the trade-off: solutions with higher ROP inevitably show greater predicted stick-slip vibration. Three representative solutions (A, B, C) are marked for further analysis.

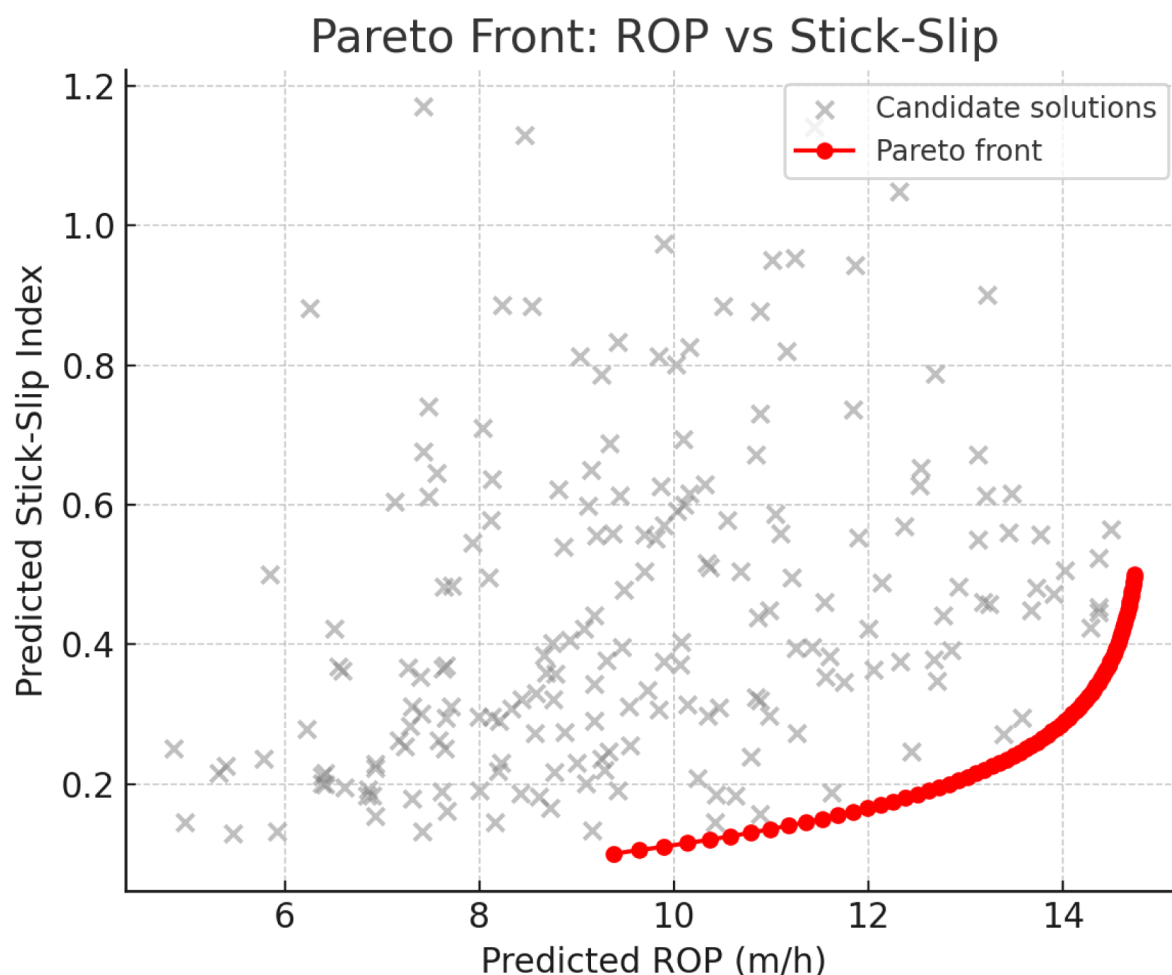


Figure 7—Pareto Front of ROP vs. Stick-Slip Index (from NSGA-II model)

At one end of the front, there is a low-vibration solution with an SSI near 0.1 (very mild stick-slip) but a lower ROP of around 9-10 m/h. At the opposite end, a high-ROP solution achieves approximately 14.5 m/h, but at the cost of an SSI ≈ 0.5 (moderate stick-slip). Between these two extremes lies a continuum of balanced solutions.

Conclusion

The drilling of EBST-26-2H demonstrated that extended-reach horizontal wells can be executed successfully in this block despite rig capacity constraints and complex downhole conditions, provided that technical risks are systematically identified and mitigated. Three primary challenges—torque and drag management, hydraulic system limitations, and drillstring dynamics issues including RSS malfunctions—were addressed through a combination of pre-planned designs and adaptive real-time measures. Solutions such as controlled tripping and reaming sequences, parameter adjustments within safe hydraulic margins, and structured RSS contingency workflows proved effective in maintaining operational continuity and avoiding severe non-productive time.

The well also marked the first application of a multi-objective drilling parameter optimization framework in the block, integrating ROP and stick-slip prediction models to support decision-making. The optimization results revealed that significant ROP improvements can be achieved while maintaining vibration severity within acceptable limits, highlighting the potential of data-driven methods in operational planning.

The engineering practices, contingency procedures, and analytical tools developed for EBST-26-2H have been incorporated into the field's technical guidelines for subsequent extended-reach and multi-well pad developments. These outcomes not only enhance operational safety and efficiency in similar geological and operational environments but also provide a replicable methodology for integrating field experience with predictive analytics in well construction projects.

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